

The reduction of OEIS A196523 modulo 3

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Abstract

OEIS A196523 is defined by the functional equation $A(x) = x + x A^4(x)$, where A^4 is the 4-th compositional iterate, and carries a conjecture of P. D. Hanna that $a(n) \equiv 1 \pmod{3}$ for every $n \geq 1$. It is true. The equation determines the coefficients one at a time, with $a(N)$ entering with coefficient 1, so it determines them modulo 3 as well. Modulo 3 the series $x/(1-x)$ satisfies the equation exactly, because its iterates are $x/(1-jx)$ and $4 \equiv 1$ modulo 3. Uniqueness then identifies the reduction. The argument is an identity between rational functions over $\mathbb{Z}/3\mathbb{Z}$; nothing is truncated or estimated.

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1 The sequence and the conjecture

OEIS A196523 is “G.f. $A(x)$ satisfies $A(x) = x + x^*A(A(A(A(x))))$.”. It has offset 1 and begins

1, 1, 4, 28, 262, 2944, 37666, ...

Writing $A(x) = \sum_{n \geq 1} a(n)x^n$ and A^i for the i -th compositional iterate of A , the entry defines A by

$$A(x) = x + x A^4(x). \quad (1)$$

This is the only input taken from the entry.

The entry separately carries the following comment:

Conjecture: $a(n) \equiv 1 \pmod{3}$ for $n > 1$. — *Paul D. Hanna*

As of the “Last modified” line on the live entry (Jun 13 2026 09:32:51, revision 30) the statement is still recorded as a conjecture, and nothing on the entry records it as settled either way.

2 The equation determines the sequence

Lemma 1. *Let $A(x) = \sum_{n \geq 1} a(n)x^n$ with $a(1) = 1$. Then for every $i \geq 1$ and every $N \geq 1$,*

$$[x^N]A^i = i a(N) + (a \text{ polynomial in } a(1), \dots, a(N-1)).$$

Proof. Induction on i . For $i = 1$ the statement is $[x^N]A = a(N)$. Suppose it holds for i and put $B = A^i$, so $[x^1]B = 1$ and $[x^N]B = i a(N) + \dots$. Then $A^{i+1} = A \circ B = \sum_{m \geq 1} a(m)B^m$, so

$$[x^N]A^{i+1} = \sum_{m=1}^N a(m) [x^N]B^m.$$

The term $m = N$ contributes $a(N)([x^1]B)^N = a(N)$; the term $m = 1$ contributes $[x^N]B = i a(N) + \dots$; and for $2 \leq m \leq N-1$ every monomial of $[x^N]B^m$ is a product of $m \geq 2$ coefficients of B whose indices sum to N , so each index is at most $N-1$ and no $a(N)$ occurs. Adding up gives $(i+1)a(N)$ plus terms in $a(1), \dots, a(N-1)$. $\square$

Lemma 2. *There is exactly one sequence of integers $a(1), a(2), \dots$ satisfying (1), and for each $N \geq 2$ the equation read at order x^N has the form $a(N) = (a \text{ polynomial with integer coefficients in } a(1), \dots, a(N-1))$.*

Proof. Comparing coefficients of x in (1) gives $a(1) = 1$, since $x A^4(x)$ has no linear term. Let $N \geq 2$. The left side contributes $a(N)$. On the right, $[x^N](x A^4) = [x^{N-1}]A^4$, which by Lemma 1 involves only $a(1), \dots, a(N-1)$. So the coefficient of $a(N)$ in the order- x^N equation is 1, the equation solves for $a(N)$ over $\mathbb{Z}$, and induction gives existence and uniqueness. $\square$

Corollary 3. *Let $m \geq 2$. If $\bar{A} \in (\mathbb{Z}/m\mathbb{Z})[[x]]$ satisfies $\bar{A} = x + O(x^2)$ and (1) with all coefficients read modulo m , then $a(n) \equiv [x^n]\bar{A} \pmod{m}$ for every n .*

Proof. Reduction modulo m is a ring homomorphism, so it carries the integer recursion of Lemma 2 to the same recursion over $\mathbb{Z}/m\mathbb{Z}$, in which the coefficient of $a(N)$ is still 1 and hence still invertible. That recursion has a unique solution, and both $a \pmod{m}$ and the coefficients of $\bar{A}$ solve it. $\square$

3 The reduction modulo 3

Write $f(x) = \frac{x}{1-x}$ over $\mathbb{Z}/3\mathbb{Z}$.

Lemma 4. $f^i(x) = \frac{x}{1-ix}$ for every $i \geq 1$.

Proof. True for $i = 1$. If $f^i(x) = x/(1-ix)$ then

$$f^{i+1}(x) = f(f^i(x)) = \frac{\frac{x}{1-ix}}{1 - \frac{x}{1-ix}} = \frac{x}{(1-ix) - x} = \frac{x}{1-(i+1)x}.$$

$\square$

Lemma 5. *Over $\mathbb{Z}/3\mathbb{Z}$, f satisfies $f(x) = x + x f^4(x)$.*

Proof. By Lemma 4, $f^4(x) = \frac{x}{1-4x}$. Since $4 \equiv 1$ modulo 3 — this is the whole content of the choice of modulus, as $3 = 4 - 1$ — we have $1 - 4x = 1 - x$ in $(\mathbb{Z}/3\mathbb{Z})[x]$, so $f^4(x) = \frac{x}{1-x}$. Therefore

$$x + x f^4(x) = x + \frac{x^2}{1-x} = \frac{x(1-x) + x^2}{1-x} = \frac{x}{1-x} = f(x).$$

$\square$

Theorem 6. *Over $\mathbb{Z}/3\mathbb{Z}$ the reduction of A is $x/(1-x)$; equivalently $a(n) \equiv 1 \pmod{3}$ for every $n \geq 1$. This is the conjecture of Section 1.*

Proof. By Lemma 5 the series $f = x/(1-x)$ satisfies the reduced equation, and $f = x + O(x^2)$. By Corollary 3 the reduction of A equals f , every coefficient of which is 1. $\square$

Remark 7. The proof used only $4 \equiv 1$ modulo 3. The same argument shows that for every $i \geq 2$ the sequence defined by $A(x) = x + x A^i(x)$ has all of its terms congruent to 1 modulo $i-1$. It also shows where the argument stops: if $m \nmid i-1$ then $1-ix$ and $1-x$ differ in $(\mathbb{Z}/m\mathbb{Z})[x]$, so $x/(1-x)$ is not a solution of the reduced equation and this route says nothing modulo m .

4 Verification

Three checks, each independent of the algebra above.

First, the recursion of Lemma 2 was run in exact rational arithmetic and its output compared against the entry's DATA at every one of the 20 published indices; the values agree exactly, and every one is an integer. A recursion that reproduced only some of them would mean the functional equation had been transcribed wrongly.

Second, at every step the coefficient of $a(N)$ was computed rather than assumed, by evaluating the order- x^N equation at $a(N) = 0$ and at $a(N) = 1$ and subtracting; it came out 1 at every N up to 23, as Lemma 2 requires. The run was continued to $n = 23$ and every term reduced modulo 3; all residues are 1.

Third, both sides of (1) were expanded independently as formal power series over $\mathbb{Z}/3\mathbb{Z}$ to order 30, starting from the coefficients of $x/(1-x)$ rather than from the closed form, and they agree at every order; the iterates f^i were confirmed to have coefficients i^{n-1} for $i \leq 6$.

References

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